

Arboretum Eco-Quest

Read the living collection and solve the checkpoints before time runs out.

Win condition: Most accurate checkpoint solutions with safe field conduct.

THE SCIENCE YOU ARE USING

Observation as evidence. Identifying a tree is detective work: leaf shape, bark texture, branching pattern, and seeds are all clues. A dichotomous key turns those clues into a series of either-or choices that lead to a name.

Ecosystems in place. The arboretum is a living classroom of ecosystems. Each checkpoint asks you to observe carefully, classify what you see, and record the evidence that justifies your answer.

HOW TO BUILD AND RUN IT

- 1 Get the map and rules.** Each team gets a route map and the field-conduct rules. Stay on paths, stay together.
 - 2 Solve each checkpoint.** Use leaf, bark, and seed clues with the key to answer each station.
 - 3 Collect evidence tokens.** Record the specific clue that justifies each answer and collect the token.
 - 4 Keep an efficient pace.** Order your route to avoid backtracking; pace counts but accuracy counts more.
 - 5 Check your answers.** Before returning, confirm each answer cites real evidence, not a guess.
- Safety:** Stay on route. No off-trail shortcuts. Buddy system and hydration check. Allergy: Pollen and insect-sting awareness.

MATERIALS

Clue cards	set
Route map	per team
Clipboards	6
Pencils	shared
Evidence token bags	per team
QR codes	optional

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why):

The ONE variable we are testing:

Baseline result:

After redesign:

DEFEND YOUR DESIGN

What evidence made you change your design?

If you ran it again, what is the ONE thing you would change?

HOW YOU SCORE (100 POINTS)

Scoring criterion	Points
Checkpoint accuracy	50
Evidence quality	20
Pace	20
Teamwork and conduct	10
Total	100